SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE:CSA1105

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 6 **Total Marks:30**

Annexure A

Scenario-Based

Code of Conduct:

I P.Malreddy (192372015) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
P.Malreddy

Annexure A

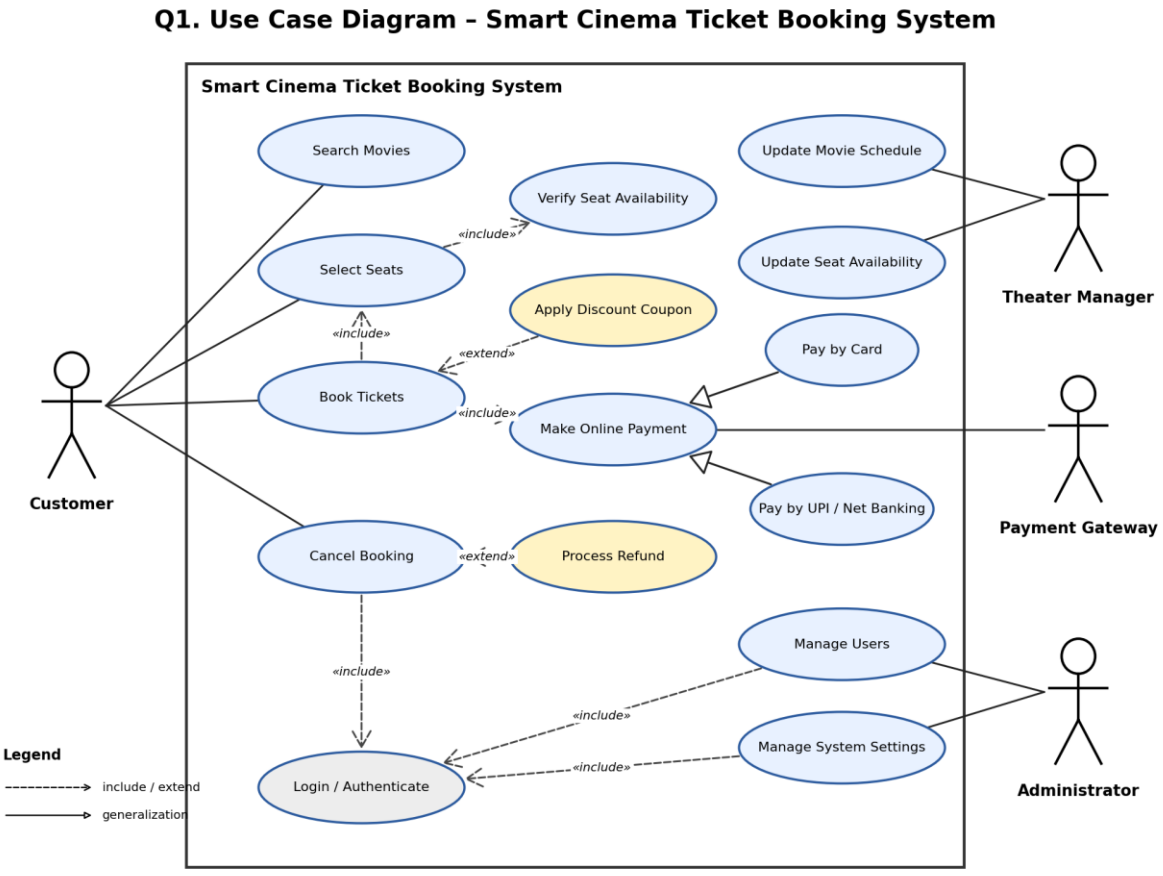
Scenario-Based

1. Use Case Diagram Scenario

A **Smart Cinema Ticket Booking System** allows customers to search movies, select seats, book tickets, cancel bookings, and make online payments. Theater managers update movie schedules and seat availability, while administrators manage users and system settings. The interactions between actors and system functions are not clearly defined.

Based on this scenario, analyze the system requirements, identify all actors and use cases, and design a **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to ensure complete system functionality.

(5 Marks)

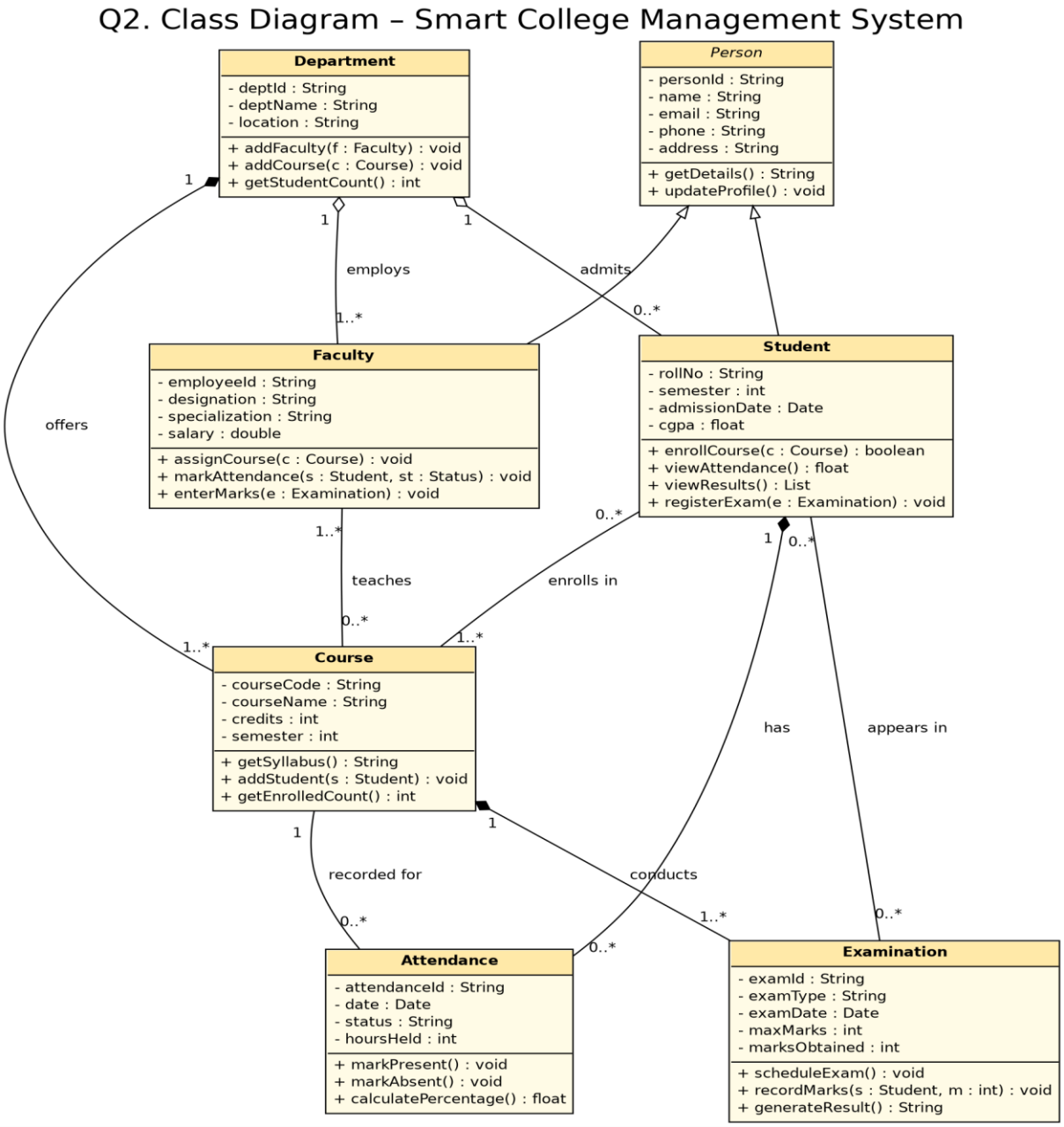


2. Class Diagram Scenario

A **Smart College Management System** consists of entities such as Student, Faculty, Department, Course, Attendance, and Examination. The existing design does not clearly define class attributes, operations, or relationships among these entities.

Based on this scenario, identify the required classes, define suitable attributes and methods, establish relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a detailed **Class Diagram**.

(5 Marks)

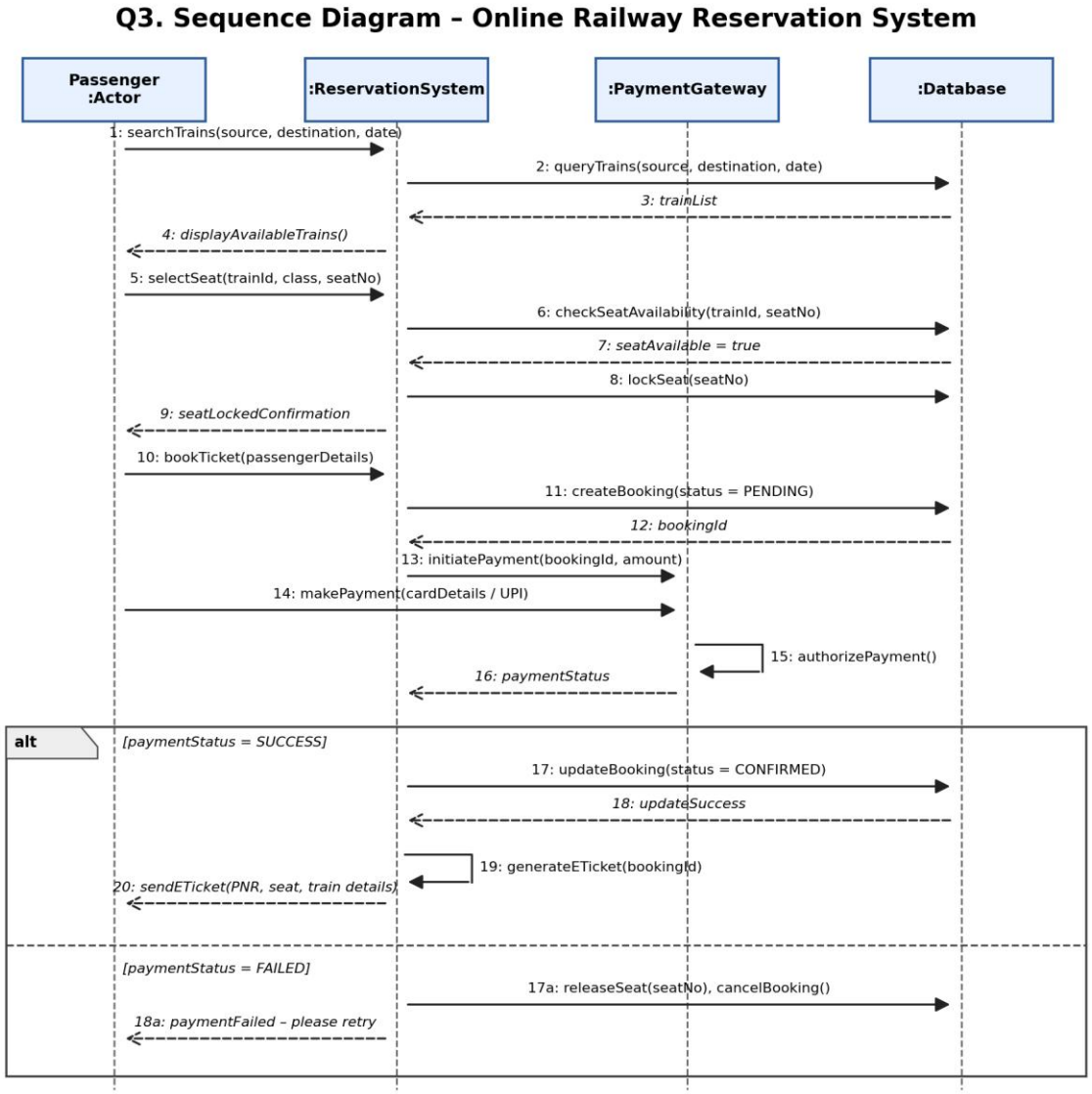


3. Sequence Diagram Scenario

In an **Online Railway Reservation System**, a passenger searches for trains, selects a seat, books a ticket, makes payment, and receives an e-ticket. The communication among the passenger, reservation system, payment gateway, and database is incomplete.

Based on this scenario, analyze the sequence of operations, identify participating objects and message flows, and design a **Sequence Diagram** representing the complete interaction process.

(5 Marks)

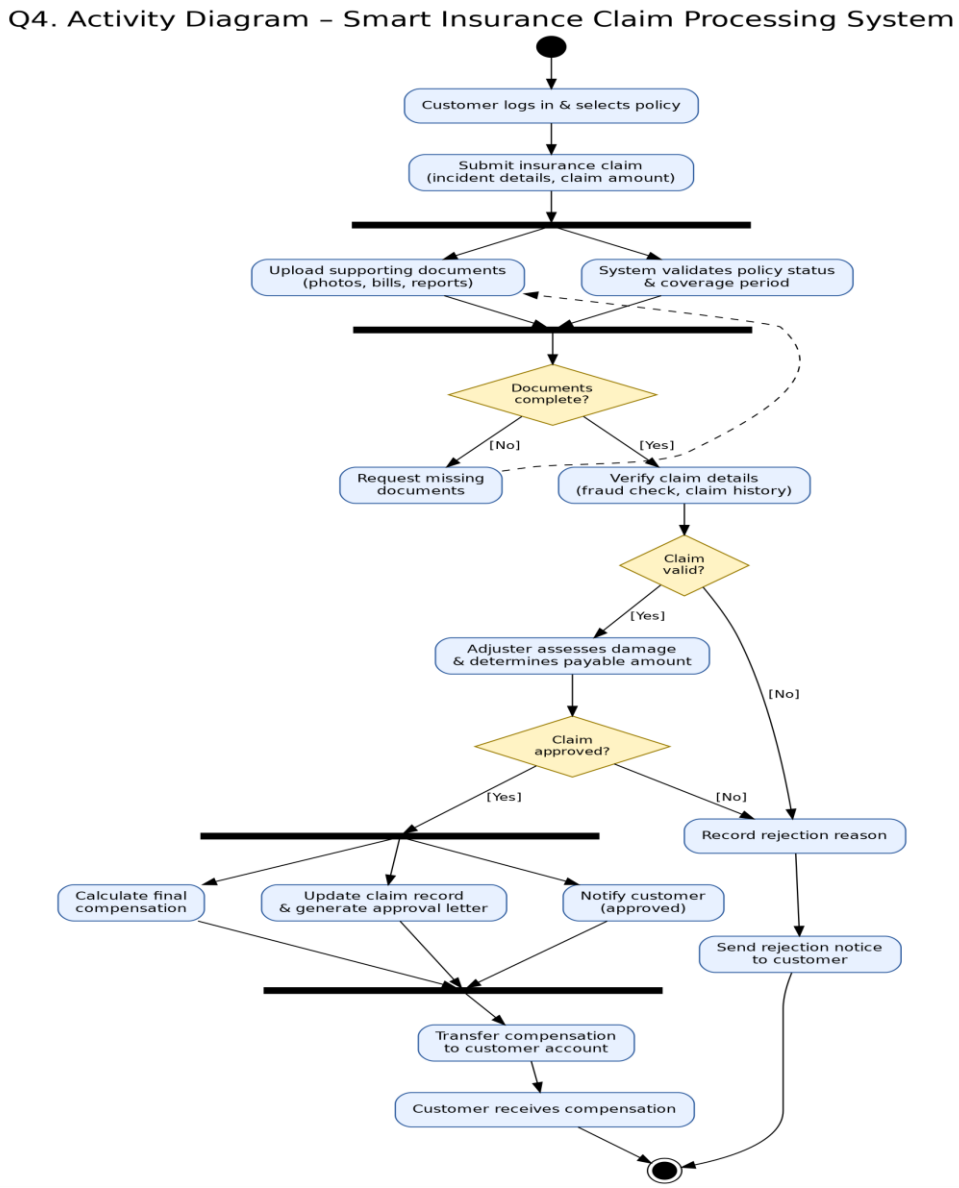


4. Activity Diagram Scenario

A **Smart Insurance Claim Processing System** allows customers to submit insurance claims, upload supporting documents, verify claim details, approve or reject claims, and receive compensation. The workflow does not clearly represent decision-making or concurrent activities.

Based on this scenario, analyze the workflow, identify decision points and parallel activities, and design an optimized **Activity Diagram** using decision nodes, forks, and joins.

(5 Marks)

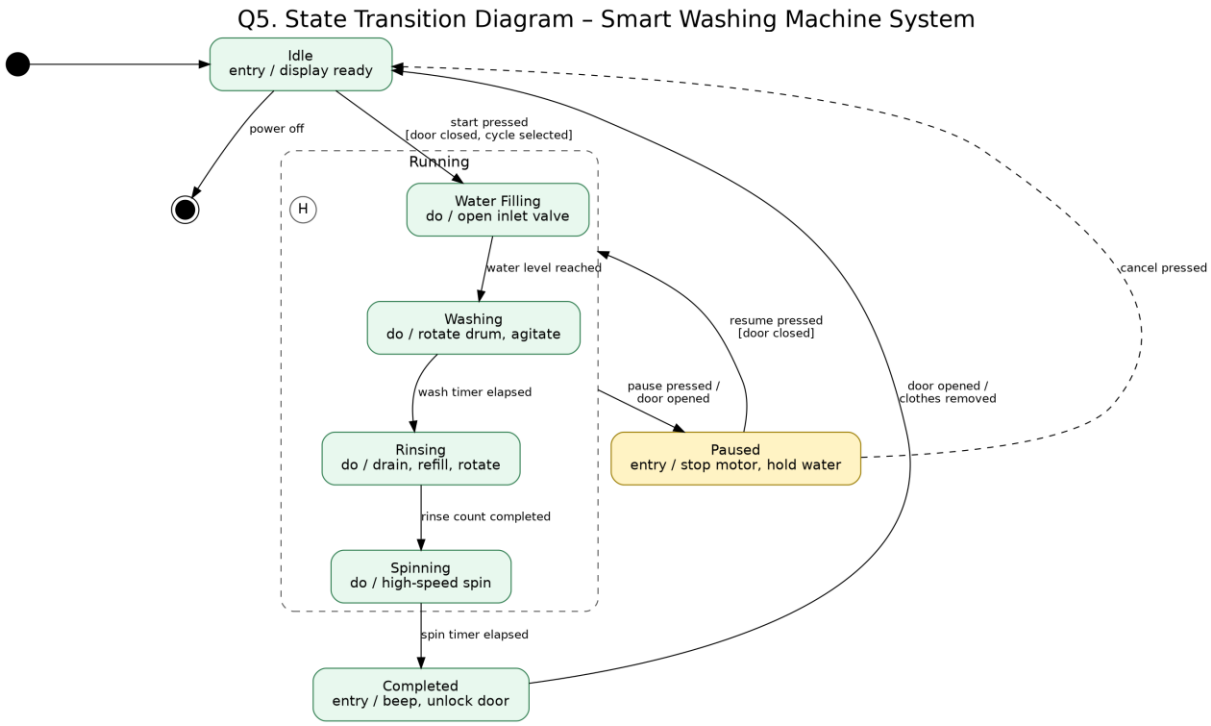


5. State Diagram Scenario

A **Smart Washing Machine System** operates through states such as Idle, Water Filling, Washing, Rinsing, Spinning, Paused, and Completed. The transitions between these operational states are not clearly specified.

Based on this scenario, identify all possible states, triggering events, and transitions, and construct a **State Transition Diagram** that accurately represents the washing machine's behavior.

(5 Marks)



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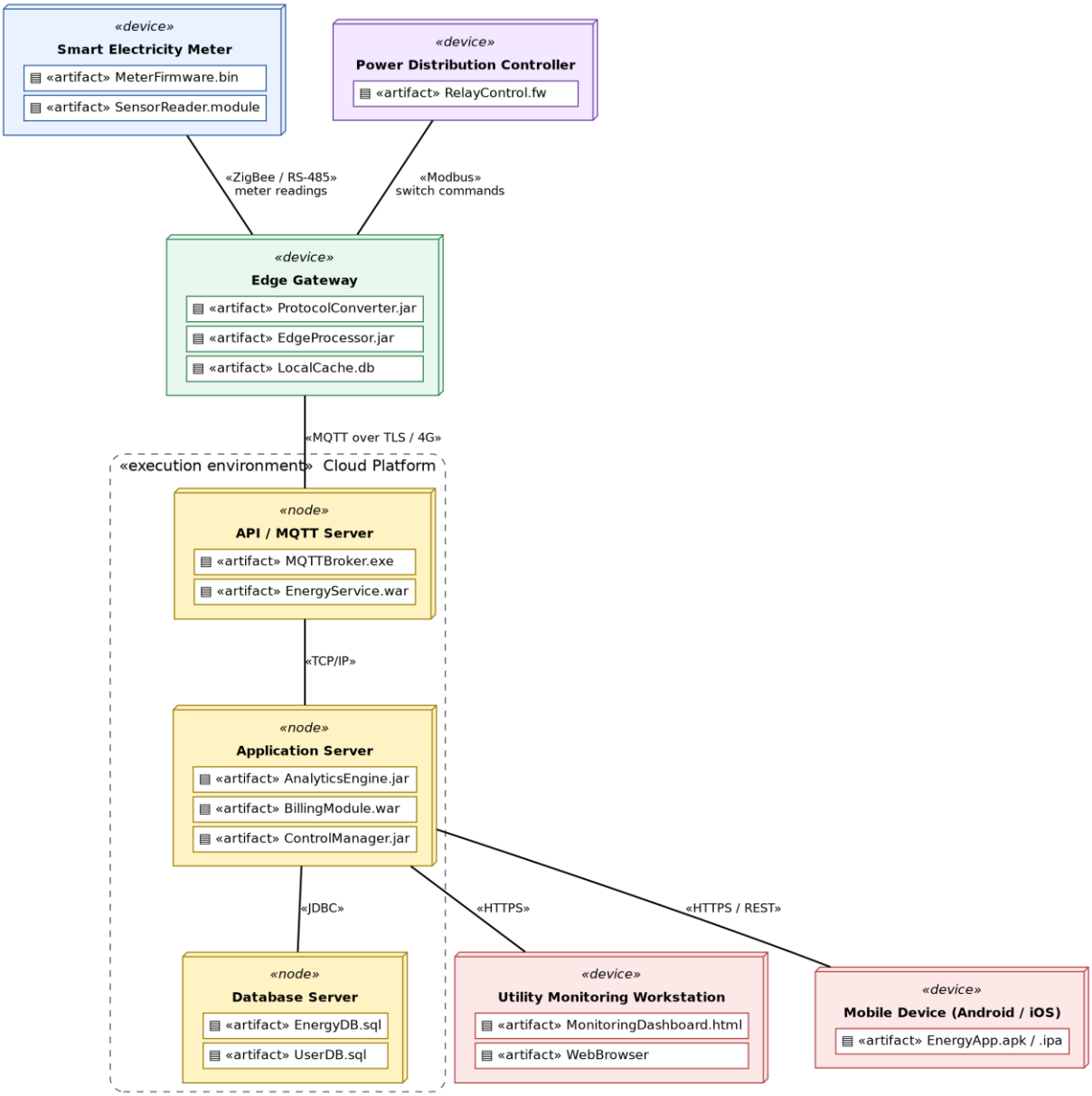
6. Deployment Diagram Scenario

A **Smart Energy Management System** uses smart electricity meters, edge gateways, cloud servers, utility monitoring dashboards, and mobile applications to monitor energy consumption and control power distribution. The deployment architecture does not clearly illustrate hardware nodes, deployed software artifacts, or communication links.

Based on this scenario, analyze the deployment environment, identify all hardware nodes and software artifacts, and design a **Deployment Diagram** showing the physical architecture and communication links.

(5 Marks)

Q6. Deployment Diagram – Smart Energy Management System



RUBRICS FOR CALCULATION

Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand